Coils, kinetic plasma and a maintained fusion state

A physics specification and mathematical research programme for PlasmaLean

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The problem. Construct external coils and an operating plasma that produces useful fusion power while remaining confined and within prescribed temperature, density, field and wall-load limits. It is enough to succeed for one design and a specified neighbourhood of initial data and manufacturing errors. It is not necessary to solve the unrestricted global regularity problem for every plasma.

The proposed route. Use geometry to construct a good magnetic background; compute a self-consistent plasma in that field; then prove estimates for perturbations that are *not* required to share the background symmetry. The natural deterministic model for turbulence is a kinetic equation coupled to electromagnetic fields. The practical core calculation is usually nonlinear gyrokinetics, coupled to profile evolution and edge/exhaust modelling.

What this brief supplies. A nonrelativistic multispecies kinetic–Maxwell parent system; explicit Coulomb, fusion, radiation, neutral and boundary interfaces; coil and circuit equations; exact energy accounting; a concrete gyrokinetic pilot; and proof targets connecting geometric design to heat loss. It distinguishes established equations, conditional deductions and speculative methods. No plasma physicist's endorsement is claimed, and no operating parameter set or stable burning solution is constructed here.

Three distinctions that must survive every formalization.

1. A vacuum field with good particle orbits is not yet a self-consistent finite-pressure plasma.
2. Smoothness is not stability, stability is not confinement, and confinement is not a favourable heating balance.
3. A theorem for a specified PDE is different from evidence that this PDE predicts the relevant experiment accurately.

Immediate recommendation. Preserve PlasmaLean's existing geometric and thermal work. Add a common candidate/data contract and a kinetic-to-transport flux interface. First establish a verified finite operating interval with explicit error margins. An all-time stable driven state is a subsequent, stronger target. Do not make a complete microscopic derivation from 10^{23} particles a prerequisite.

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1 Scope, geometry and unknowns

The principal case is a classical, weakly coupled, deuterium–tritium magnetic-confinement plasma. The electromagnetic field is self-consistent. The Coulomb interaction is represented by collective fields plus the Landau collision operator. This is a standard kinetic approximation, not an exact N-body description. Its applicability and the accuracy of each subsequent reduction must be checked for the chosen regime.

Use SI units; temperatures T_s are energies in joules, with electronvolts used only after conversion. A reported scalar kinetic temperature is one third of the pressure-tensor trace divided by density. It need not describe the full distribution.

Distinguish the *material vessel* $\Omega_{\mathrm{v}} \subset \mathbb{R}^3$, a selected hot-core region Ω_{c}, a coil region Ω_{coil} outside the vessel, and a field domain $\mathcal{D}$ containing them. A magnetic flux surface inside the vessel is not a reflecting material wall. The physical metric is the Euclidean metric inherited from the actual embedding.

Unknown or input	Meaning
$f_s(x, v, t) \geq 0$	Number distribution for electrons, deuterons, tritons, helium nuclei and selected impurity charge states. One helium distribution can contain both born energetic alphas and thermal ash. Separate populations require a conservative transfer rule.
f_a^0, f_n	Neutral-atom/molecule distributions and a neutron distribution where needed. Neutral charge states may instead be included in a unified species list.
$\mathbf{E}, \mathbf{B}$	Total resolved electromagnetic fields, including coil, plasma, wall and imposed antenna contributions.
$I_k(t)$, γ_k	Coil currents and closed coil paths, or finite-cross-section current shapes. Coil paths are design variables, normally fixed during a discharge.
$I_{\nu,p}(x, \omega, t)$	Radiation intensity at frequency ν, direction ω, polarization/mode p, when radiation is not replaced by a justified optically thin closure.
Wall/circuit variables	Wall temperature, charge and induced currents; prescribed power-supply/control laws.
Physical data	Cross-sections, atomic levels and rates, Coulomb logarithms, source spectra, wall response, material laws, coil tolerances and boundary data.

The nonrelativistic equations below require $T_e/(m_e c^2) \ll 1$ and negligible dynamically important relativistic tails. Their errors must be assessed rather than declared zero. If necessary, use momentum p, velocity $v_s = p/\sqrt{m_s^2 + |p|^2/c^2}$, relativistic kinetic energy and the relativistic collision/radiation operators *together*. Replacing only v in a nonrelativistic Landau model is inconsistent. Nuclear-product energies should use consistent reaction kinematics.

An operating claim must specify the time interval, admissible initial perturbations, controls and uncertainties. For example:

$$T_{i,\min} \leq T_D, T_T \leq T_{i,\max}, \quad T_{e,\min} \leq T_e \leq T_{e,\max}, \quad n_D, n_T \geq n_{\min} \quad \text{on } \Omega_{\mathrm{c}} \times [0, \tau].$$

Add permitted particle losses, alpha deposition, wall loads, plasma gain and coil constraints. Species temperatures, density profiles and nonthermal tails cannot all be replaced by one scalar temperature.

2 Coils and the electromagnetic equations

2.1 A field must be generated by an admissible current

For a filament design with closed smooth curves $\gamma_k : S^1 \to \mathbb{R}^3$ and currents I_k, the static vacuum field away from the wires is

$$\mathbf{B}_{\mathrm{coil}}(x) = \frac{\mu_0}{4\pi} \sum_k I_k \int_{S^1} \frac{\gamma_k'(\theta) \times [x - \gamma_k(\theta)]}{|x - \gamma_k(\theta)|^3} \, \mathrm{d}\theta. \tag{2.1}$$

It is divergence-free and curl-free in a current-free plasma region. Filament curves are an idealization; final verification must include finite conductor cross-sections and every return path. A return conductor at an astronomical distance is not an engineering realization.

A finite-volume representation is

$$\mathbf{j}_{\mathrm{coil}}(x,t) = \sum_k I_k(t)\, \mathbf{w}_k(x), \qquad \nabla \cdot \mathbf{w}_k = 0, \qquad \int_{\Sigma_k} \mathbf{w}_k \cdot \mathbf{n} \, \mathrm{d}S = N_k, \tag{2.2}$$

where $\mathbf{w}_k$ has units m^{-2}, is supported in the actual closed winding pack, and N_k is the number of turns. The static Biot–Savart integral with $\mathbf{j}_{\mathrm{coil}}$ replaces (2.1). Minimum coil–plasma and coil–coil distances, curvature/bend limits, conductor current density, peak conductor field, forces and tolerances are separate constraints. Existing precise-QS coil designs and direct coil optimization provide relevant comparators [10, 11].

For slow current changes, a circuit description can be coupled to the field:

$$V_k^{\mathrm{drive}} = R_k(T_k) I_k + \frac{d}{dt} \Phi_k^{\mathrm{link}}, \qquad \Phi_k^{\mathrm{link}} = \sum_l L_{kl} I_l + \Phi_{k,\mathrm{plasma}} + \Phi_{k,\mathrm{wall}}. \tag{2.3}$$

The linkage includes turns and mutual inductances. Finite cross-sections regularize self-inductance. Specify V_k^{drive} or a control law; a fixed-current design is an explicitly controlled special case. The coil temperature T_k must be supplied by a stated cooling/material model or evolved with its thermal balance. For superconductors, negligible resistance is an operating approximation subject to critical-current, field, temperature and protection limits. Do not add both $L\dot{I}$ and the derivative of a flux that already includes LI.

Equation (2.3) is a magnetoquasistatic circuit reduction. If relevant current frequencies make retardation/capacitance important, resolve conductor electrodynamics instead. An instantaneous time-dependent Biot–Savart field is not an exact solution of the full Maxwell system.

2.2 Full resolved Maxwell system

In $\mathcal{D}$, with plasma distributions extended only through their physical domains,

$$\partial_t \mathbf{B} = -\nabla \times \mathbf{E}, \qquad \nabla \cdot \mathbf{B} = 0, \tag{2.4}$$

$$\epsilon_0 \partial_t \mathbf{E} = \mu_0^{-1} \nabla \times \mathbf{B} - \mathbf{j}_{\mathrm{tot}}, \qquad \epsilon_0 \nabla \cdot \mathbf{E} = \rho_{\mathrm{tot}}, \tag{2.5}$$

$$\rho_{\mathrm{pl}} = \sum_s q_s \int_{\mathbb{R}^3} f_s \, \mathrm{d}v, \qquad \mathbf{j}_{\mathrm{pl}} = \sum_s q_s \int_{\mathbb{R}^3} v f_s \, \mathrm{d}v, \tag{2.6}$$

$$\mathbf{j}_{\mathrm{tot}} = \mathbf{j}_{\mathrm{pl}} + \mathbf{j}_{\mathrm{coil}} + \mathbf{j}_{\mathrm{wall}} + \mathbf{j}_{\mathrm{antenna}}. \tag{2.7}$$

Include the corresponding charges in ρ_{tot}, including surface charge. For a stationary ordinary conductor, a simple material law is $\mathbf{j}_{\mathrm{wall}} = \sigma_{\mathrm{wall}} \mathbf{E}$ in its volume, supplemented by charge continuity and interface conditions. More detailed conductor laws can replace it.

Initial Gauss constraints and compatible boundary data are mandatory. Charge consistency gives

$$\partial_t \rho_{\mathrm{tot}} + \nabla\cdot \mathbf{j}_{\mathrm{tot}} = 0, \qquad \partial_t(\nabla\cdot \mathbf{B}) = 0, \qquad \partial_t(\epsilon_0 \nabla\cdot \mathbf{E} - \rho_{\mathrm{tot}}) = 0. \tag{2.8}$$

The last two equalities follow from the evolution equations and charge continuity. Particle injection must be charge-balanced or accompanied by its electrode/source current. A source term creating uncompensated charge is not repaired by choosing a numerical divergence cleaner.

One may write $\mathbf{B} = \mathbf{B}_{\mathrm{coil}} + \mathbf{B}_{\mathrm{response}}$ for bookkeeping, but evolve or solve the *total* field consistently. At finite pressure, plasma currents can change surfaces, transform and quasisymmetry. Zero *net toroidal* plasma current does not mean $\mathbf{j}_{\mathrm{pl}} = 0$, nor does it exclude every local current-driven instability.

3 The multispecies kinetic parent system

3.1 Evolution and the Coulomb collision operator

For each charged species, the nonrelativistic parent equation is

$$\begin{aligned} \partial_t f_s + v\cdot\nabla_x f_s + \frac{q_s}{m_s}(\mathbf{E} + v\times\mathbf{B})\cdot\nabla_v f_s = {} & \sum_{s'} C_{ss'}[f_s, f_{s'}] \\ & + \mathcal{R}_s^{\mathrm{fus}} + \mathcal{R}_s^{\mathrm{atom}} + \mathcal{R}_s^{\gamma} + \mathcal{H}_s + \mathcal{S}_s. \end{aligned} \tag{3.1}$$

Here $\mathcal{H}_s$ is a specified unresolved heating operator, if used, and $\mathcal{S}_s$ is a specified fuel/beam/source term. The electromagnetic force is nonlinear through (2.4)–(2.7); turbulent fluctuations are part of the solution.

For number distributions, define

$$U(z) = \frac{1}{|z|}\left(I - \frac{z\otimes z}{|z|^2}\right), \qquad \Gamma_{ss'} = \frac{q_s^2 q_{s'}^2 \ln\Lambda_{ss'}}{8\pi\epsilon_0^2}.$$

With a symmetric positive Coulomb logarithm at each (x,t), the convention used here is

$$C_{ss'} = \frac{\Gamma_{ss'}}{m_s}\nabla_v \cdot \int_{\mathbb{R}^3} U(v-v')\left[\frac{f_{s'}(v')}{m_s}\nabla_v f_s(v) - \frac{f_s(v)}{m_{s'}}\nabla_{v'} f_{s'}(v')\right] \mathrm{d}v'. \tag{3.2}$$

The kernel is singular at $z = 0$; use a specified weak or regular solution framework. This normalization agrees with the multispecies convention in [1]. A velocity-dependent modification of $\ln\Lambda$ needs a new conservation/entropy check.

For sufficiently regular decaying distributions, pairwise symmetrization gives

$$\int C_{ss'}\,\mathrm{d}v = 0, \tag{3.3}$$

$$\sum_s \int m_s v \sum_{s'} C_{ss'}\,\mathrm{d}v = 0, \qquad \sum_s \int \tfrac{1}{2} m_s |v|^2 \sum_{s'} C_{ss'}\,\mathrm{d}v = 0. \tag{3.4}$$

The collision contribution to $H = \sum_s \int f_s \log f_s \,\mathrm{d}x\,\mathrm{d}v$ is given below. A fixed reference phase-space density is understood inside each logarithm; number conservation makes this choice irrelevant to the collision derivative.

$$\begin{aligned} \left.\frac{dH}{dt}\right|_{\mathrm{coll}} &= -\frac{1}{2}\sum_{s,s'}\int \Gamma_{ss'} f_s f'_{s'}\, a_{ss'}\cdot U(v-v') a_{ss'}\,\mathrm{d}x\,\mathrm{d}v\,\mathrm{d}v' \le 0, \\ a_{ss'} &= \frac{\nabla_v \log f_s}{m_s} - \frac{\nabla_{v'}\log f'_{s'}}{m_{s'}}. \end{aligned} \tag{3.5}$$

These are local collision identities, not monotonicity of a driven reactor with sources and walls. Collisions exchange energy between species and convert nonthermal structure into heat; they do not remove total particle energy.

The field and collision equations retain parallel phase mixing, resonances, magnetic drifts, trapping, finite gyroradius physics and nonlinear electromagnetic interactions within this kinetic approximation. Adding an empirical turbulent diffusivity to the same fully resolved kinetic calculation would generally double-count transport.

3.2 Fusion reactions and alpha deposition

For the dominant reaction $D + T \to \alpha + n$, define $g = |v - w|$ and a differential cross-section $d\sigma_{DT}/d\omega$. The local reaction rate is

$$R_{DT}(x,t) = \iint f_D(x,v,t) f_T(x,w,t)\, g\, \sigma_{DT}(g)\, \mathrm{d}v\, \mathrm{d}w. \tag{3.6}$$

For example,

$$\mathcal{R}_D^{\mathrm{fus}}(v) = -f_D(v) \int f_T(w) g \sigma_{DT}(g)\, \mathrm{d}w,$$

with the analogous tritium sink. A precise and compact definition of all four reaction terms is their weak form: for arbitrary test functions ζ_s,

$$\begin{aligned} &\sum_{s=D,T,\alpha,n} \int \zeta_s(v) \mathcal{R}_s^{\mathrm{fus}}(v)\, \mathrm{d}v \\ &\quad = \iiint f_D(v) f_T(w) g \frac{d\sigma_{DT}}{d\omega} \left[\zeta_\alpha(v'_\alpha) + \zeta_n(v'_n) - \zeta_D(v) - \zeta_T(w)\right] \mathrm{d}\omega\, \mathrm{d}w\, \mathrm{d}v. \end{aligned} \tag{3.7}$$

Product velocities are fixed by the reaction angular law and consistent momentum/energy kinematics. The number-source moments are $(-R_{DT}, -R_{DT}, R_{DT}, R_{DT})$, charge is conserved, and the increase of kinetic energy is $Q_{DT} R_{DT}$ when nuclear rest-energy depletion is excluded from the state energy. Here $Q_{DT} \simeq 17.6$ MeV. The familiar 3.5 MeV alpha and 14.1 MeV neutron energies are approximate low-reactant-energy values, not fixed product energies for every incoming pair.

Use cross-section data, such as Bosch–Hale within its specified validity ranges, with documented corrections and interpolation error [7]. The shortcut $R_{DT} = n_D n_T \langle \sigma v \rangle (T_i)$ requires the relevant Maxwellian assumptions. It is not a definition valid for an arbitrary kinetic state. Add D–D and other channels if a quantified estimate shows their relevance.

Evolve alphas with (3.1); their Coulomb slowing and escape then determine deposition. The energy delivered to another species by alphas is

$$P_{s \leftarrow \alpha}(x,t) = \int \tfrac{1}{2} m_s |v|^2 C_{s\alpha}[f_s, f_\alpha]\, \mathrm{d}v. \tag{3.8}$$

Do not simultaneously add all of 3.5 MeV R_{DT} as local thermal heating: some birth energy remains in fast alphas or is lost before thermalization. A reduced deposition kernel is acceptable only with a stated approximation/error bound.

Neutrons have $q_n = 0$ and obey a transport equation

$$\partial_t f_n + v \cdot \nabla_x f_n = \mathcal{R}_n^{\mathrm{fus}} + \mathcal{R}_n^{\mathrm{matter}}.$$

For a core-plasma calculation their escape can be treated as a measured outgoing energy channel; blanket transport is needed for deposited wall/blanket power and plant performance.

3.3 Heating, fueling and atomic reactions

Choose one consistent representation of each heating mechanism. Resolve an antenna wave in Maxwell and its absorption kinetically, *or* use a derived quasilinear heating operator with a specified wave spectrum. Do not count its absorbed power a second time. Neutral beams and pellets need particle, momentum and energy source spectra; balanced electron/ion creation must respect (2.8).

Ionization, recombination, charge exchange, excitation and de-excitation can be specified by reaction kernels analogous to (3.7), now with internal atomic energies and photons included in the energy ledger. This is a definition of a family of models until the kernels, species and levels are supplied. Maxwellian-averaged rate tables are not automatically valid for a strongly non-Maxwellian electron tail. OPEN-ADAS and the EIRENE databases provide relevant atomic data and model infrastructure [8, 9].

4 Radiation, neutrals and physical boundaries

4.1 Radiation without energy double counting

For incoherent radiation, a transport description is

$$\frac{1}{c}\partial_t I_{\nu,p} + \omega \cdot \nabla_x I_{\nu,p} = j_{\nu,p}[f] - \kappa_{\nu,p}[f] I_{\nu,p} + \mathcal{C}^{\mathrm{sc}}_{\nu,p}[I, f]. \tag{4.1}$$

The coefficients have the intensity convention appropriate to this equation. Define

$$u_\gamma = \frac{1}{c}\sum_p \int I_{\nu,p}\,\mathrm{d}\nu\,\mathrm{d}\omega, \qquad \mathbf{F}_\gamma = \sum_p \int \omega I_{\nu,p}\,\mathrm{d}\nu\,\mathrm{d}\omega.$$

The matter operators $\mathcal{R}^\gamma_a$ and the atomic internal-energy terms must remove precisely the energy gained by radiation. Here a ranges over all participating material species, including neutral atoms and their internal states:

$$\sum_a \int \tfrac{1}{2} m_a |v|^2 \mathcal{R}^\gamma_a \,\mathrm{d}v + P^\gamma_{\mathrm{internal}} = -\sum_p \int \left(j_{\nu,p} - \kappa_{\nu,p} I_{\nu,p} + \mathcal{C}^{\mathrm{sc}}_{\nu,p}\right) \mathrm{d}\nu\,\mathrm{d}\omega. \tag{4.2}$$

Emission, absorption and scattering should come from common kernels, not independently chosen terms satisfying only an energy moment. Assign each channel once: a radiative atomic event contributes its matter term to either $\mathcal{R}^{\mathrm{atom}}$ or $\mathcal{R}^\gamma$, with the same event supplying the photon equation; it is never added twice. In the ledger above, $\mathcal{R}^\gamma_a$ denotes all such radiative matter contributions, whichever bookkeeping label is used. Include bremsstrahlung, relevant line/recombination radiation and electron-cyclotron/synchrotron radiation, including wall reflection/reabsorption where material.

Equation (4.1) presupposes a radiation-transfer regime. Polarization and refraction may require mode-dependent rays or a more complete wave model. Declare which electromagnetic scales are resolved in Maxwell and which are in radiation transfer; the same radiation must not be counted in both Poynting flux and a separate radiative loss.

In a justified optically thin, near-Maxwellian limit, replace the radiation block by a documented power closure, schematically

$$P_{\mathrm{rad}} = n_e \sum_{Z,z} n_{Z,z} L_{Z,z}(T_e, n_e) + P_{\mathrm{cyc}}.$$

The chosen cooling coefficients must specify whether they already contain bremsstrahlung and recombination. Use exact dataset versions, temperature ranges and units; some tabulations use $\mathrm{W\,cm^3}$ rather than $\mathrm{W\,m^3}$. An unspecified constant impurity fraction is a modelling assumption, not a solved impurity balance.

4.2 Neutrals, wall losses and exhaust

For a neutral species,

$$\partial_t f_a^0 + v \cdot \nabla_x f_a^0 = \mathcal{R}_a^{\mathrm{atom}}[f, f^0, I] + \mathcal{R}_a^{\gamma}[f, f^0, I] + \mathcal{S}_a^0. \tag{4.3}$$

Kinetic neutral transport and coupled plasma/neutral edge models are standard tools for recycling, impurity penetration and divertor heat/particle fluxes [9]. Using Monte Carlo to solve them does not make the underlying distribution equations stochastic.

At a material boundary, let $\mathbf{n}$ point out of the plasma and $\Gamma_\pm = \{(x, v) : \pm v \cdot \mathbf{n} > 0\}$. A general deterministic inflow law is

$$|v \cdot \mathbf{n}| f_s|_{\Gamma_-}(v) = \mathcal{F}_s^{\mathrm{inj}}(v) + \sum_a \int_{v' \cdot \mathbf{n} > 0} K_{s \leftarrow a}(x, v; v', T_w)\, f_a(v')\, v' \cdot \mathbf{n}\, \mathrm{d}v'. \tag{4.4}$$

The kernel includes specified reflection, recycling, secondary emission and sputtering. Its particle and energy yields need not all be one. Wall charge, deposited energy and emitted material must balance its moments.

A perfectly absorbing boundary with $K = 0$ is a legitimate stated benchmark. A specular reflecting boundary is also a mathematical benchmark but suppresses physical wall loss and cannot establish realistic exhaust performance. At a plasma–vacuum computational interface that is *not* a wall, use transmission/matching, not (4.4).

Resolve the Debye sheath with an appropriate kinetic/electrostatic model or use a justified sheath boundary law. Quasineutral core gyrokinetics does not itself resolve the sheath. The wall electric potential or floating-current condition and surface-charge evolution must be compatible with Maxwell. For orientation from region 1 to 2, ordinary interface laws include

$$\mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0, \quad \mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = 0, \quad \epsilon_0 \mathbf{n} \cdot (\mathbf{E}_2 - \mathbf{E}_1) = \sigma_\Sigma, \quad \mathbf{n} \times (\mathbf{B}_2 - \mathbf{B}_1) = \mu_0 \mathbf{K}_\Sigma.$$

A perfect conductor has zero tangential electric field; it does not require the pre-existing normal magnetic flux to be zero.

The outer field problem also needs a choice: Maxwell on all of space with the specified incoming fields and an outgoing radiation condition for the scattered field, or a finite computational boundary with a documented transparent/absorbing approximation and reflection error. Incoming photon intensity must be prescribed; at material surfaces use the chosen emission and reflection kernel. A numerical boundary must transmit the intended coil field and cannot arbitrarily set its normal component to zero.

If wall temperature is a claimed output, supply material heat transport, for example

$$\rho_w c_w(T_w) \partial_t T_w = \nabla \cdot (k_w(T_w) \nabla T_w) + P_{\mathrm{Joule}} + P_{\mathrm{vol}},$$

with plasma, neutral, photon and neutron deposition at the appropriate surfaces/volumes and a specified coolant boundary law. This is a thermal interface, not a mechanical or neutron-damage qualification.

5 Energy, heat flux and the actual confinement claim

5.1 Exact moments of the parent kinetic equation

For each species define

$$n_s = \int f_s \,\mathrm{d}v, \qquad n_s \mathbf{u}_s = \int v f_s \,\mathrm{d}v, \tag{5.1}$$

$$\Pi_s = m_s \int (v - \mathbf{u}_s) \otimes (v - \mathbf{u}_s) f_s \,\mathrm{d}v, \qquad T_s = \frac{\operatorname{tr} \Pi_s}{3 n_s}, \tag{5.2}$$

$$K_s = \int \tfrac{1}{2} m_s |v|^2 f_s \,\mathrm{d}v, \qquad \mathbf{F}_s = \int \tfrac{1}{2} m_s |v|^2 v f_s \,\mathrm{d}v, \tag{5.3}$$

$$\mathbf{q}_s = \int \tfrac{1}{2} m_s |v - \mathbf{u}_s|^2 (v - \mathbf{u}_s) f_s \,\mathrm{d}v. \tag{5.4}$$

The exact decomposition is

$$\mathbf{F}_s = \left(\tfrac{1}{2} m_s n_s |\mathbf{u}_s|^2 + \tfrac{3}{2} n_s T_s \right) \mathbf{u}_s + \Pi_s \mathbf{u}_s + \mathbf{q}_s. \tag{5.5}$$

Thus an energy flux and a conductive heat flux are different quantities. Neither is determined by scalar temperature alone.

Writing $\mathcal{C}_s$ for the entire right side of (3.1),

$$\partial_t n_s + \nabla \cdot (n_s \mathbf{u}_s) = \int \mathcal{C}_s \,\mathrm{d}v, \tag{5.6}$$

$$\partial_t (m_s n_s \mathbf{u}_s) + \nabla \cdot (m_s n_s \mathbf{u}_s \otimes \mathbf{u}_s + \Pi_s) = q_s n_s (\mathbf{E} + \mathbf{u}_s \times \mathbf{B}) + \int m_s v \mathcal{C}_s \,\mathrm{d}v, \tag{5.7}$$

$$\partial_t K_s + \nabla \cdot \mathbf{F}_s = \mathbf{j}_s \cdot \mathbf{E} + \int \tfrac{1}{2} m_s |v|^2 \mathcal{C}_s \,\mathrm{d}v. \tag{5.8}$$

These identities require the indicated moments and integration by parts to exist. They supply the coupling and diagnostics without an independent fluid closure.

For the resolved fields,

$$\partial_t \left(\frac{\epsilon_0 |\mathbf{E}|^2}{2} + \frac{|\mathbf{B}|^2}{2\mu_0} \right) + \nabla \cdot \left(\frac{\mathbf{E} \times \mathbf{B}}{\mu_0} \right) = -\mathbf{j}_{\mathrm{tot}} \cdot \mathbf{E}. \tag{5.9}$$

Adding (5.8) over species cancels plasma electric work with its part of (5.9). Landau energy exchange cancels between species. Adding radiation cancels (4.2). Nuclear rest-energy conversion, external sources, conductor work and boundary fluxes remain. This is the first decisive consistency check for a coupled implementation.

Do not add a second Ohmic heating term to a kinetic energy equation that already contains the same electric work. Do not add circuit magnetic energy to a field-energy integral that already includes that coil field. Turbulence redistributes energy and can transfer it between fields and particles; it is not an unaccounted destruction term in this total-energy balance.

5.2 From a kinetic flux to a transport equation

On fixed nested reference surfaces $\psi = \text{const}$, set

$$V'(\psi) = \int_{\psi=\text{const}} \frac{\mathrm{d}S}{|\nabla \psi|}, \qquad \langle a \rangle_\psi = \frac{1}{V'} \int_{\psi=\text{const}} \frac{a}{|\nabla \psi|} \,\mathrm{d}S.$$

Then, for smooth fluxes,

$$\partial_t \langle K_s \rangle_\psi + \frac{1}{V'} \partial_\psi \left[V' \langle \mathbf{F}_s \cdot \nabla \psi \rangle_\psi \right] = \left\langle \mathbf{j}_s \cdot \mathbf{E} + \int \tfrac{1}{2} m_s |v|^2 \mathcal{C}_s \,\mathrm{d}v \right\rangle_\psi. \tag{5.10}$$

This is a divergence identity, not the assumption that the solution is symmetric. Moving surfaces require Reynolds-transport terms. Islands or stochastic field regions require a different coordinate treatment; do not continue to use nonexistent global flux coordinates.

For low mean flow and suitable scale separation, (5.10) yields slow particle and thermal-profile equations

$$\partial_t \bar{n}_s + \frac{1}{V'}\partial_\psi(V'\Gamma_s) = S_s^n, \tag{5.11}$$

$$\partial_t(\tfrac{3}{2}\bar{n}_s\bar{T}_s) + \frac{1}{V'}\partial_\psi(V'\mathcal{Q}_s) = S_s^E. \tag{5.12}$$

All omitted mean-flow, stress-work and geometry terms must be explicitly accounted for in a derivation. Turbulent and neoclassical fluxes are outputs of a consistent split of the same kinetic model. A multiscale derivation of this general organization is given in [6]; its specific rotating-tokamak equilibrium equations must not be transplanted unchanged to a stellarator.

In the leading electrostatic gyrokinetic approximation, a typical turbulent energy-flux contribution has the structure

$$\mathcal{Q}_s^{\mathrm{turb}} = \left\langle \int \mathcal{E}_s h_s\, \bar{\mathbf{v}}_{E,s}\cdot\nabla\psi\, \mathrm{d}^3v \right\rangle_{\psi,\mathrm{fast}}, \qquad \mathcal{E}_s = \tfrac{1}{2}m_s v_\parallel^2 + \mu B_0. \tag{5.13}$$

Use the proper gyroaverages and phase-space measure. Electromagnetic fluctuations also give magnetic-flutter transport and consistent field-energy terms. This illustrative formula is not the complete electromagnetic flux. A time/spatial average here is a deterministic operation, not stochastic forcing.

PlasmaLean's supplied diffusivity model can remain a reduced module, but it needs a proved or explicitly empirical bridge from these fluxes. Turbulent transport need not be local, purely diffusive or described by a positive scalar $\chi(T)$. A heat-flux bound alone does not control its spatial divergence. To preserve a pointwise temperature barrier, bound the actual operator defect, use a monotone conservative finite-volume comparison with justified face-flux dependence, or bound the kinetic state strongly enough to control local temperature.

5.3 Observable bounds that are actually needed

For any fixed core Ω_{c}, integrating (5.8) gives its energy budget with the signed surface flux $\int_{\partial\Omega_{\mathrm{c}}}\mathbf{F}_s\cdot\mathbf{n}$. Particle, neutral, radiation and field channels must have a common ledger. Integral core energy does not imply a pointwise hot core.

For a simple illustration of a necessary norm choice, if

$$\sup_x \left\| \langle v\rangle^\ell (f_s - \widehat{f}_s)\right\|_{L^2_v} \le r_s, \qquad \ell > 9/2,$$

then Cauchy–Schwarz bounds the errors in moments through order three by explicit constants times r_s. Here $\langle v\rangle$ uses a fixed reference speed to nondimensionalize v. With lower density bounds, temperature is a locally Lipschitz function of its zeroth, first and second moments. Obtaining this norm from spatial Sobolev control requires a proved embedding; a volume L^2 estimate alone also does not control wall traces. Strong enough trace estimates or direct boundary-flux certificates are indispensable.

Temperature bounds alone do not give a fusion-rate lower bound for arbitrary distributions. If the reaction kernel $k(v,w) = g\sigma_{DT}(g)$ is bounded by a verified $K_{\max}$, then

$$\begin{aligned}|R_{DT}(f_D,f_T) - R_{DT}(F_D,F_T)| \le K_{\max}\big[\, &\|f_D - F_D\|_{L^1_v}\,\|f_T\|_{L^1_v} \\ &+ \|F_D\|_{L^1_v}\,\|f_T - F_T\|_{L^1_v}\,\big].\end{aligned} \tag{5.14}$$

This follows by adding and subtracting $F_D f_T$. If the kernel is bounded only on a truncated velocity domain, add separately certified tail terms. Equation (5.14) is one concrete bridge from a Maxwellian reactivity calculation to a genuinely kinetic yield claim.

For a maintained thermal state, require deposited alpha heating plus permitted auxiliary heating to balance transport and radiation, while maintaining fuel and removing ash. Scientific plasma gain $Q = P_{\mathrm{fus}}/P_{\mathrm{aux}}$ and net electric plant gain are different: coil, cryogenic, pumping and conversion powers enter the latter [22]. Arbitrarily large externally supplied heating is not evidence of a useful fusion power source.

6 The practical turbulence calculation

6.1 A hierarchy with explicit validity checks

The parent system above specifies the physical ledger. Directly resolving every cyclotron period, Debye length, radiation frequency and burn timescale is generally an unsuitable computational programme. A useful implementation combines:

1. A coil-generated magnetic equilibrium, updated for finite-pressure plasma response.
2. Nonlinear electromagnetic gyrokinetics for core fluctuations, with kinetic electrons, relevant ions, collisions and the actual geometry.
3. Slow density, temperature and flow evolution using the resulting fluxes, plus consistent neoclassical transport and alpha deposition.
4. Edge, neutral, impurity and sheath models, with matching fluxes and documented errors.

Gyrokinetics is an asymptotic reduction, not a symmetry restriction to axisymmetric perturbations. Typical orderings include $\rho_s/L \ll 1$, $\omega/\Omega_s \ll 1$ and $k_\parallel/k_\perp \ll 1$, while $k_\perp \rho_s$ can be order one. For a local delta-f model also require small fluctuations and slowly varying background profiles. Near the edge, near narrow barriers, for large global modes or energetic-particle effects, those assumptions need separate tests. A conserved reduced energy requires consistent polarization and magnetization, not a hand-assembled kinetic equation plus arbitrary field closures [3].

6.2 An explicit electrostatic pilot, not the final reactor model

For a first proof/certificate exercise, choose a fixed background $\mathbf{B}_0 = B_0\mathbf{b}$, no strong equilibrium flow, locally Maxwellian thermal species

$$F_{0s} = n_s \left(\frac{m_s}{2\pi T_s}\right)^{3/2} e^{-\mathcal{E}_s/T_s}, \qquad \mu = \frac{m_s v_\perp^2}{2B_0}, \quad \mathcal{E}_s = \tfrac{1}{2} m_s v_\parallel^2 + \mu B_0.$$

Use the standard split $\delta f_s = -q_s \phi F_{0s}/T_s + h_s$, with h_s evaluated at the appropriate gyrocentre when reconstructing particle space. Write $\bar{\phi}_s = \langle \phi \rangle_R$ and

$$\bar{\mathbf{v}}_{E,s} = \frac{\mathbf{b} \times \nabla \bar{\phi}_s}{B_0}, \tag{6.1}$$

$$\mathbf{v}_{M,s} = \frac{\mathbf{b} \times [\mu \nabla B_0 + m_s v_\parallel^2 (\mathbf{b} \cdot \nabla)\mathbf{b}]}{q_s B_0}. \tag{6.2}$$

The leading local nonlinear electrostatic equations are

$$\begin{aligned} &\partial_t h_s + (v_\parallel \mathbf{b} + \mathbf{v}_{M,s} + \bar{\mathbf{v}}_{E,s}) \cdot \nabla h_s - \frac{\mu}{m_s} \mathbf{b} \cdot \nabla B_0 \, \partial_{v_\parallel} h_s \\ &\quad = \frac{q_s F_{0s}}{T_s} \partial_t \bar{\phi}_s - \bar{\mathbf{v}}_{E,s} \cdot \nabla\big|_{\mathcal{E}} F_{0s} + C_s^{\mathrm{GK}}[h], \end{aligned} \tag{6.3}$$

$$\sum_s \frac{n_s q_s^2}{T_s}\phi = \sum_s q_s \int \langle h_s \rangle_x \, \mathrm{d}^3 v. \tag{6.4}$$

Here $\langle \cdot \rangle_x$ is the inverse gyrocentre-to-particle averaging at fixed particle position. With gyrotropic velocity integration, $\mathrm{d}^3 v = 2\pi B_0 \, \mathrm{d}v_\parallel \, \mathrm{d}\mu / m_s$ at this order. The collision term is the consistently gyroaveraged linearized multispecies operator, or a named conserving approximation with quantified consequences. Specify perpendicular periodicity, the field-line end matching, gauge/mean-potential convention, initial data and profile gradients. Polarization is encoded by the distribution split and gyroaverages; replacing (6.4) by equality of uncorrected gyrocentre densities loses it.

This is a local *fluctuation* equation after the background neoclassical response has been separated at the chosen order; it does not declare F_{0s} an exact full-system equilibrium. A global or full-f model must retain the corresponding background drift/source balance and the consistent phase-space Jacobian. The pilot follows the structure used in standard local codes [4, 5]. It retains nonlinear $E \times B$ advection. Keep zonal modes, which can regulate turbulence, and test kinetic electron dynamics rather than using an adiabatic-electron closure for every heat channel. A constant-gradient flux-tube calculation is not a whole-device temperature-maintenance proof.

For electromagnetic calculations add the vector potential, parallel electric acceleration and magnetic fluctuations through a single consistent published gyrokinetic formulation. For example, the parallel Ampere equation has the local leading form

$$-\nabla_\perp^2 A_\parallel = \mu_0 \sum_s q_s \int v_\parallel \langle h_s \rangle_x \, \mathrm{d}^3 v,$$

with the matching Hamiltonian, perpendicular magnetic response, polarization and current definitions from that same formulation. This equation alone is *not* the electromagnetic extension of (6.3). The electromagnetic parent evolution remains (2.4)–(3.1); implementation of a reduced electromagnetic system must pin the source equations and their ordering [3, 4].

The mean radial electric field and its shear also need a self-consistent balance or a controlled supplied profile. Near exact QS, the usual neoclassical ambipolarity condition can become degenerate; it must not be assumed to uniquely determine the field. Fast-alpha distributions generally are not local Maxwellians and require their appropriate kinetic treatment.

6.3 Evidence required before accepting a small heat loss

Use resolution and domain-size studies in both position and velocity, electron/ion scale checks where relevant, electromagnetic sensitivity, collision-model sensitivity, zonal-flow treatment and a total-energy balance. Inspect artificial dissipation and velocity-space tails. Compare independent codes on a published benchmark before applying a new certificate to a design. Compare global and local calculations when radial nonlocality or variation around the surface matters. Finite averaging times give sampling uncertainty in a flux estimate even for a deterministic calculation.

QS, maximum-J geometry and available-energy metrics are useful screening tools. The trapped-electron available-energy study [17] reports a correlation with saturated transport in its test class; that is not a universal rigorous upper bound on reactor heat flux. Ion-temperature-gradient, electron-temperature-gradient, electromagnetic and energetic-particle channels require their own checks. An energy reservoir has units of joules and a loss rate has units of watts; converting one to the other needs dynamical information.

7 Quasisymmetry and a geometric ansatz

7.1 What the symmetry means

If suitable magnetic surfaces and Boozer coordinates have been established, ordinary QS is expressed by

$$|\mathbf{B}_0|(\psi,\theta,\zeta) = \mathcal{B}(\psi, M\theta - N\zeta). \tag{7.1}$$

It is a condition on the field strength in special magnetic coordinates, not invariance of the vector field under an ordinary Euclidean rotation. The coordinates, their periods and their relation to the physical field are part of the claim. Approximate QS must be measured throughout the relevant volume with a specified norm, not just on one surface. Exact, strong and weak QS statements have distinct hypotheses and guiding-centre consequences [13].

For a vacuum field a useful strong geometric target uses $\alpha = \mathbf{B}^\flat$, $\beta = \iota_{\mathbf{B}}\mathrm{vol}_3$ and a generator $\mathbf{u}$:

$$d\alpha = 0, \quad d\beta = 0, \quad \iota_{\mathbf{u}}\beta = d\Psi, \quad \mathcal{L}_{\mathbf{u}}\alpha = 0, \quad \mathcal{L}_{\mathbf{u}}\mathrm{vol}_3 = 0. \tag{7.2}$$

The convention here gives $\mathbf{B}\times\mathbf{u} = \nabla\Psi$. On a connected vacuum region, $\mathcal{L}_{\mathbf{u}}\alpha = 0$ implies $\mathbf{B}\cdot\mathbf{u} = C$ constant. Then

$$\mathbf{u} = \frac{\nabla\Psi\times\mathbf{B} + C\mathbf{B}}{|\mathbf{B}|^2}, \qquad \mathbf{B}\cdot\nabla\Psi = 0.$$

This reduces part of the search to a first-order system for (Ψ, C) once $\mathbf{B}$ is fixed. Smooth extension across an axis, nonvanishing, nested surfaces, marked rotational transform and a periodic generator remain global requirements. These equations are a deliberately strong target, not an assertion that every practical approximate-QS stellarator must satisfy them exactly.

Precise approximate QS with coils has already been obtained numerically [10]. Thus an exact strong-QS existence theorem should be pursued as a potentially useful mathematical route, not made an unnecessary prerequisite for practical design. Field-line confinement and guiding-centre invariants do not guarantee confinement of every finite-gyroradius particle or collective stability.

7.2 Respect the prescribed physical domain

The earlier fixed-domain formulation is retained: if a solid torus Ω is given as a *magnetic domain*, with

$$d\alpha = 0, \quad d{*_3}\alpha = 0, \quad \alpha(\mathbf{n}) = 0, \quad \oint_\gamma \alpha = P$$

for a generator γ of $H_1(\Omega)$, then the normalized tangent harmonic field is unique. Indeed the difference of two solutions has zero periods, hence is $d\varphi$; harmonicity and the homogeneous Neumann condition imply $\int_\Omega |\nabla\varphi|^2 = 0$. Equivalently the harmonic Neumann space has dimension $b_1(\Omega) = 1$.

There is no hidden family of physical fields available by changing a holomorphic bundle metric while holding all these data fixed. One must test the unique field, or explicitly declare a larger design problem allowing the magnetic boundary to vary. A fixed *vessel* with unspecified magnetic normal field is a different problem and does not have this uniqueness conclusion. Coils must realize the selected boundary-value solution, not an unrelated one.

7.3 A concrete complex-geometric representation

On the auxiliary Euclidean product $X = \Omega\times\mathbb{R}_\tau$, take orientation $d\tau\wedge\mathrm{vol}_3$ and set

$$F = d\tau\wedge\alpha - {*_3}\alpha. \tag{7.3}$$

Then

$$*_4F = -F, \qquad dF = 0 \quad \Longleftrightarrow \quad d\alpha = 0,\ d*_3\alpha = 0$$

for τ-independent α. With the compatible standard Kähler structure, for example $z_1 = \tau + ix_1$, $z_2 = x_2 + ix_3$, this is a primitive $(1,1)$ form; iF can be represented as an Abelian unitary curvature. On a solid-torus product with $\mathbb{R}$, $H^2(X) = 0$, so there is no additional curvature-integrality obstruction. Periods, boundary data and holonomy still matter.

This calculation packages *vacuum Maxwell* as an Abelian anti-self-duality/Hermitian–Yang–Mills problem. It does not supply QS, coils or finite-pressure kinetic stability. The auxiliary coordinate τ is not physical time. Changing the Kähler metric changes the Hodge star and generally changes the physical problem.

Useful directions are: derive compatibility/prolongation conditions for (7.2); exploit holomorphic strip estimates to sharpen coil/KAM certificates; and seek boundary integral or moment-map formulations that retain the exact Euclidean reconstruction. A higher-rank KYM or vortex ansatz earns its place only by giving a proved map back to (2.1), (7.2) and the fixed physical geometry. Merely importing the equations from a different product surface does not do that.

A related coil construction uses a winding-surface current potential:

$$\mathbf{K} = \mathbf{n}_\Sigma \times \nabla_\Sigma \Phi, \qquad \Phi = \Phi_{\text{single}} + \frac{I_{\text{pol}}}{2\pi}\theta + \frac{I_{\text{tor}}}{2\pi}\zeta.$$

The two affine coefficients encode the toroidal periods; their identification with physical current components must follow the chosen angle/orientation convention. REGCOIL is an established regularized current-potential method [12]. Hodge theory is natural here, but discretizing the current sheet into finite coils still needs an error estimate in norms controlling surfaces and transport.

7.4 A plasma ansatz is a reference, not the entire perturbation class

An initial reference can have flux-function $n_s(\psi)$ and $T_s(\psi)$, a mean electric potential and a small neoclassical correction. Alternatively, a collisionless reduced model may admit $F_s = \mathscr{F}_s(H_s, \mu_s, P_{\text{QS},s})$ when its claimed invariants have actually been proved. These invariants generally belong to a particular guiding-centre approximation, not the complete collisional, driven, finite-gyroradius system.

There is a simple warning against treating a hot-profile Maxwellian as an exact equilibrium. In a source-free stationary full kinetic model with zero flow, local Maxwellians and vanishing collision terms, the Lorentz magnetic force annihilates the isotropic Maxwellian. The remaining equation contains

$$v \cdot \left[\nabla \log n_s + \left(\frac{m_s|v|^2}{2T_s} - \frac{3}{2}\right)\nabla \log T_s - \frac{q_s \mathbf{E}}{T_s}\right] = 0.$$

Its cubic coefficient forces $\nabla T_s = 0$, and then $\nabla \log n_s = q_s\mathbf{E}/T_s$. A nonconstant hot-core Maxwellian is therefore a leading-order reference, not an exact zero-flow equilibrium of this full source-free equation. Corrections, transport and sources must be solved. Cross-species collisional equilibrium also requires a common temperature and mean velocity.

Likewise, in ordinary static isotropic MHD with negligible electric force, $\mathbf{j}_{\text{pl}} \times \mathbf{B} = \nabla p$. Taking the *total* field to be vacuum forces constant pressure. The external vacuum field can coexist with pressure only because the plasma response is included, or because one has explicitly controlled a low-beta approximation.

Use

$$f_s = \widehat{f}_s + g_s, \qquad \mathbf{E} = \widehat{\mathbf{E}} + \delta\mathbf{E}, \qquad \mathbf{B} = \widehat{\mathbf{B}} + \delta\mathbf{B}$$

with unrestricted admissible three-dimensional perturbations. Prove the equations for the reference and every perturbation component, including those transverse to the ansatz. Exact symmetry of a selected solution provides no information about those modes.

8 What the proposed analytic ideas can realistically do

8.1 BKM and kinetic continuation: a conditional bridge

For ideal constant-density incompressible MHD, in Alfvén units $\mathbf{h} = \mathbf{B}/\sqrt{\mu_0\rho_0}$, a BKM-type continuation condition involves

$$\int_0^T \left(\|\nabla\times \mathbf{u}\|_{L^\infty} + \|\nabla\times \mathbf{h}\|_{L^\infty} \right) \mathrm{d}t < \infty. \tag{8.1}$$

The Caflisch–Klapper–Steele theorem has its own periodic-domain and regularity hypotheses [18]. It does not say this integral is automatically finite. It does not apply unchanged to compressible, thermal, kinetic or wall-bounded equations. Nor is finiteness of the integral over the entire infinite half-line required to continue through every finite time.

For kinetic models the analogous quantities involve field control, velocity moments/support and derivatives of distributions. Luk–Strain provide continuation criteria for the *collisionless relativistic* Vlasov–Maxwell system [19]; these are methodological precedents, not theorems for (3.1) with Coulomb collisions, radiation and a vessel. Coulomb diffusion creates velocity tails, so compact velocity support is particularly unsuitable as a casually imported hypothesis.

The useful task is to derive a continuation theorem for one chosen model and norm, then produce that theorem's hypotheses by a quantitative estimate. A smooth solution can still cool or lose all its fuel; continuation must be coupled to the observable bounds of Section 5.

8.2 Entropy and hypocoercivity: the strongest first analytic route

For a positive reference state $\widehat{f}_s$, a candidate starting functional is

$$\begin{aligned}\mathscr{L}_0 = T_0 \sum_s \int \left[f_s \log \frac{f_s}{\widehat{f}_s} - f_s + \widehat{f}_s \right] \mathrm{d}x\,\mathrm{d}v \\ + \frac{\epsilon_0}{2}\int |\mathbf{E}-\widehat{\mathbf{E}}|^2\,\mathrm{d}x + \frac{1}{2\mu_0}\int |\mathbf{B}-\widehat{\mathbf{B}}|^2\,\mathrm{d}x,\end{aligned} \tag{8.2}$$

where $T_0 > 0$ is a fixed energy scale. This is nonnegative; no dissipative inequality for it is asserted merely from that fact. Different species reference temperatures, spatial gradients, reference residuals, driving and boundaries all contribute terms to its derivative.

For the electrostatic pilot, a more directly usable quadratic functional is available. In its local perpendicular Fourier representation, set

$$\eta_{s,k} = h_{s,k} - \frac{q_s F_{0s}}{T_s} J_{0s}\phi_k, \quad J_{0s} = J_0(k_\perp v_\perp/\Omega_s), \quad b_s = \frac{k_\perp^2 T_s}{m_s\Omega_s^2}, \quad \Gamma_{0s} = I_0(b_s)e^{-b_s},$$

where $\Omega_s = |q_s|B_0/m_s$. Quasineutrality gives the equivalent forms

$$\begin{aligned}W_k &= \frac{1}{2}\left\langle \sum_s \left[T_s \int \frac{|h_{s,k}|^2}{F_{0s}}\,\mathrm{d}^3v - \frac{n_s q_s^2}{T_s}|\phi_k|^2 \right] \right\rangle_{\text{tube}} \\ &= \frac{1}{2}\left\langle \sum_s \left[T_s \int \frac{|\eta_{s,k}|^2}{F_{0s}}\,\mathrm{d}^3v + \frac{n_s q_s^2}{T_s}(1-\Gamma_{0s})|\phi_k|^2 \right] \right\rangle_{\text{tube}} \geq 0.\end{aligned} \tag{8.3}$$

Here J_0, I_0 are the ordinary and modified Bessel functions and the tube average uses the compatible flux-tube volume measure. This is the electrostatic specialization of the standard free energy, with a factor-of-two normalization choice [14, 16]. Fix the constant-potential gauge and treat conserved/neutral modes explicitly; positivity alone is not a uniform coercivity or derivative bound.

The local gyrokinetic free-energy balance for $W = \sum_k W_k$ has the organizing form

$$\frac{dW}{dt} = P_{\text{gradients}} + P_{\text{sources}} - D_{\text{collisions}} - F_{\text{boundary}}. \tag{8.4}$$

With compatible closed/periodic boundaries the nonlinear gyrokinetic advection redistributes this free energy. Maintained density and temperature gradients replenish it. A nonnegative collision dissipation requires the matching collision-equilibrium or reduced-operator hypotheses; otherwise keep the reference residual and interspecies relaxation contributions explicitly. Helander–Plunk obtain rigorous bounds on linear and nonlinear free-energy growth within local gyrokinetics [14, 15, 16]. Those bounds are not automatically decay estimates or upper bounds on saturated heat flux.

A productive analogue of a geometric entropy method is a *modified kinetic energy/entropy*: add justified transport–collision cross terms, field terms and weighted derivatives to control the actual modes. Prove positivity and an explicit differential inequality. Hypocoercivity supplies a mathematical framework, but Coulomb Landau does not automatically have a uniform exponential gap in the chosen space; weights and invariant modes are essential [20]. Closed homogeneous equilibrium results do not establish a driven nonuniform reactor state.

Do not impose an external Ornstein–Uhlenbeck heat bath just to obtain an easy coercive estimate and then claim fusion heating has been modelled. A bath can be a method-development benchmark. Its energy and entropy structure differs from the conservative interspecies Landau operator.

The Perelman analogy should therefore be a request for an identity with useful coercivity, not a request to rename (8.2). Ricci flow evolves a metric; the laboratory plasma has a fixed Euclidean metric and driven kinetic evolution. Positive entropy does not bound spatial derivatives, high velocity moments or boundary heat flux by itself.

8.3 A concrete fluctuation-to-flux estimate to pursue

For the electrostatic expression (5.13), Cauchy–Schwarz gives, under the stated averaging and integrability,

$$\begin{aligned} |\mathcal{Q}_s^{\text{turb}}| \le & \left(\left\langle \int T_s h_s^2 / F_{0s} \, \mathrm{d}^3 v \right\rangle_{\psi,\text{fast}} \right)^{1/2} \\ & \times \left(\left\langle \int \frac{\mathcal{E}_s^2 F_{0s}}{T_s} |\bar{\mathbf{v}}_{E,s} \cdot \nabla \psi|^2 \, \mathrm{d}^3 v \right\rangle_{\psi,\text{fast}} \right)^{1/2}. \end{aligned} \tag{8.5}$$

This is an elementary proved inequality, not yet a useful bound: the second factor contains field gradients. Derive control of both factors by an energy norm with explicit geometry, gyroradius and spectral constants. In an electromagnetic model add magnetic-flutter and field-energy fluxes. An L^2 field amplitude alone does not control arbitrary field gradients.

Research target 1. *For a fixed coil-generated candidate and the full allowed perturbation class of a specified kinetic model, prove*

$$|\mathcal{Q}_s^{\text{turb}}| \le C_{Q,s} \mathscr{W}, \qquad \dot{\mathscr{W}} \le -\lambda \mathscr{W} + C_{\text{nl}} \mathscr{W}^{3/2} + \varepsilon(t)$$

or an appropriate nonexponential replacement. Derive, rather than assume, every norm, constant, collision/null-space restriction and source term. Then propagate the resulting flux/state bounds into Section 5's thermal and yield margins.

This target may fail for a given state because its gradients genuinely drive turbulence. Failure is useful if it identifies an unstable mode, a noncoercive direction or a quantitatively excessive loss. A different valid target is bounded statistically steady turbulence with a small time-averaged outward flux; demanding that every fluctuation decay could exclude useful operation.

8.4 Wick rotation and gauge theory

Real-time Maxwell–kinetic evolution includes propagation, resonances, phase mixing, irreversible collisions and external power. A Wick-rotated elliptic equation is not equivalent to this driven dissipative initial-boundary-value problem without a theorem specifying analyticity, continuation, positivity, boundary data and error control.

Use elliptic or geometric flows to *find a reference configuration* or a positive multiplier, if helpful. Then verify the real-time equations and perturbation estimate separately. Electromagnetism has an Abelian $U(1)$ gauge field; introducing a non-Abelian bundle may be a mathematical construction tool but is not additional physical plasma dynamics. The most concrete immediate geometric tasks are (7.3), the fixed-domain strong-QS compatibility problem, winding-surface Hodge theory and sharper analytic-strip bounds.

8.5 General relativity: a substantive precedent, with limits

There is a real precedent for importing the geometric PDE method: He–Xu–Yu study nonlinear stability of three-dimensional Alfvén waves using characteristic geometry and energy arguments inspired by Christodoulou–Klainerman [21]. Their perturbative MHD setting near a background field is not a hot kinetic plasma in a compact stellarator with walls.

Useful transfers are constraint propagation as in (2.8), gauge fixing without deleting physical flux sectors, commuted characteristic estimates, null/cancellation structures and a bootstrap around a selected background. A toroidal bounded domain can remove spatial dispersive decay and introduce recurrent rays, resonances and boundary interactions. Any proposed transfer must identify how these replace the decay mechanism of the source theorem. No Einstein equation is needed to model a laboratory stellarator.

8.6 The thermal feedback must itself be stable

Even an idealized uniform temperature obeying

$$C\,\dot{T} = P_\alpha(T) + P_{\mathrm{aux}}(T) - P_{\mathrm{loss}}(T), \qquad C > 0,$$

has a thermally unstable equilibrium T_* if the derivative of the right-hand side at T_* is positive. Spatial transport, density evolution and delayed alpha deposition complicate this further. A heating equality at one temperature is therefore insufficient. Test the full coupled thermal/fueling/control linearization and nonlinear margins.

An all-time burn also requires sustained fuel supply and ash removal. With finite initial fuel and no refueling, positive fusion power bounded away from zero for all time is impossible by the particle balances. This is an elementary obstruction that a purported entropy argument must not evade.

9 What PlasmaLean currently provides

This is a source inspection of the pinned public snapshot, not a fresh compilation or an independent audit of every Lean proof. File headers and theorem hypotheses take precedence over an optimistic prose summary.

Existing component	Scope relevant to this proposal
Repository contract/status	Distinguishes current-free transform, omnigeneity, finite-beta stability, equilibrium/surfaces and thermal performance. It explicitly reports that no candidate yet satisfies the full contract. The Dommaschk results include a nested pair of certified vacuum tori and small QA defect on those tori, not a certified volumetric burning plasma.
Radial transport kernel	Uses a supplied $D(\psi, T) \geq 0$ and source $R(\psi, T)$. Classical solutions and their derivative relations are supplied hypotheses; solution existence is not asserted. The axis can be degenerate. This is suitable for conditional thermal comparison, not a derivation of turbulence.
BKM/thermal research note	Already separates continuation, symmetry, stability and thermal bounds, and proposes residual-based evolution validation. Its corrected scope restrictions should be retained.
Relative-entropy coercivity	Proves a one-dimensional thermal entropy–Fisher inequality under boundary anchoring, coefficient bounds and an upper bound on T/T_*. Decay also assumes the time-derivative relation and a solution. It is not yet a kinetic hypocoercive estimate.
Available-energy kernel	Its header explicitly says the simplified kernel has not been checked against the cited physical source. It corrects a sign mistake in the note. The bounce-derivative expression is not yet proved equal to the derivative with moving endpoints. Preserve these qualifications.
Polygon-coil result	Reports finite field-line confinement for 52 toroidal transits, conditional on field-line existence, with an idealized return conductor at $10^6 R_0$. This is useful mathematical infrastructure, not a practical finished coil set.
Coil KAM residual certificate	The inspected coarse demonstration certifies a very large strip bound (order 10^9). The accompanying status note calls it vacuous for the intended KAM threshold. Sharper complex estimates have a concrete role here.

Two amendments are especially important. First, low net toroidal current does not exclude all local current-driven modes. Second, a bound on thermal pressure by an equilibrium pressure budget is not equality of the kinetic stress with the stress used to generate that equilibrium. The same candidate must satisfy the coupled momentum/field equations or carry an explicitly bounded residual.

The new programme does not discard the earlier MHD or thermal pilots. It changes the final interpretation: a thermal barrier with an assumed transport coefficient is one module, and the kinetic turbulent-flux producer is a separate module that must actually be supplied.

10 A theorem target that can lead to a useful certificate

10.1 Start with one maintained reference and a finite horizon

Let U contain the selected distributions, fields, profile/control variables and material variables. Fix a specific model, domain, data and norm X. Given a reconstructed reference $\widehat{U}(t)$, compute the

continuum residual

$$\mathcal{R}(t) = \partial_t \widehat{U} - \mathcal{F}(\widehat{U}).$$

Boundary, initial, Maxwell-constraint, coil-geometry and velocity-tail defects belong in the certificate as well. A residual at grid points is not a continuum residual bound.

Research target 2. *Establish local well-posedness and an appropriate continuation theorem for the selected model. Derive, for $e = U - \widehat{U}$, an estimate*

$$D^+ \|e\|_X \le a(t) \|e\|_X + b(t) \|e\|_X^2 + \epsilon(t)$$

on a stated admissible set, with certified functions a, b, ϵ. Prove that the scalar comparison solution remains inside that set on $[0, \tau + \delta]$, $\delta > 0$. Infer existence and an error bound through the closed requested interval $[0, \tau]$.

Here D^+ denotes an upper right Dini derivative when needed. This is a proposed theorem architecture, not a theorem established in this brief for the reactor system. The hard producer is the PDE estimate, including its boundaries and norms; the scalar comparison lemma is the easy part.

Compose the resulting state bound with temperature/density/flux estimates and (5.14). State the separate numerical, physical-model and data uncertainties. In general, the sum of small residuals for uncoupled models is not a residual for their coupled evolution.

10.2 A stronger all-time route

For a steady or periodic driven reference, seek a genuine norm inequality

$$D^+ r \le -\lambda r + L r^2 + \epsilon, \qquad \lambda > 0, \quad L, \epsilon \ge 0. \tag{10.1}$$

If a radius $r_* > 0$ lies within the validity region and $-\lambda r_* + L r_*^2 + \epsilon \le 0$, then the scalar flow points inward at r_*. With initial error below this radius, suitable comparison and continuation give an invariant neighbourhood. For $L > 0$, the discriminant condition $\lambda^2 > 4L\epsilon$ provides a nonempty interval of inward-pointing radii. This algebra is valid; obtaining (10.1) for the physical PDE is an open task here.

Do not demand this exponential form if the actual collision/transport operator only supports weaker decay. A time-dependent comparison, a weighted nonexponential estimate or a bounded turbulent invariant set can replace it. A periodic attractor also requires removal or treatment of phase/neutral modes.

To make the neighbourhood a *hot* one, prove that its norm controls every operating observable with margins. For example, if the reference core temperature is at least $T_{\mathrm{burn}} + \Delta_T$ and its error is at most Δ_T, a lower-temperature guarantee follows. Without this norm-to-observable step, a stability theorem has not certified a burn.

11 Work assignments and completion tests for agents

The following are proposed independent work packages, not claims that parallel agents have already completed them. Each package must publish hypotheses, a source or derivation, its exact output and a counterexample/sanity test.

P0. Freeze the physical and mathematical contract

Create one data object shared by every module: physical domain/metric, coil geometry and currents, species, field normalization, profiles, sources, wall/edge laws, time horizon, tolerances and operating margins. Record which boundary is a flux surface and which is material. Each imported theorem must be labelled *applicable*, *requires adaptation*, or *research analogy*. Deliver a dependency graph of assumptions and their producers.

P1. Certify the coil-generated background

Begin with the fixed-domain harmonic problem if that domain remains prescribed. Test the selected QS condition and nonvanishing. Otherwise explicitly enlarge the design space and compare to direct coil optimization or REGCOIL. Certify the actual finite-coil field, including returns, finite thickness and manufacturing errors. Target explicit C^k and analytic-strip estimates on a common core, not only an L^2 normal-field fit. Carry these errors into KAM, transform and QS statements. A well-fitting unrelated coil set does not complete the task.

P2. Find one self-consistent finite-pressure reference

Use the same fields, pressure, currents and densities for equilibrium and transport. Include bootstrap and diamagnetic response and the radial electric field. Construct a steady, periodic or slowly evolving kinetic reference and its residual; do not declare a local Maxwellian an exact solution. Check macroscopic pressure and current modes as well as microturbulence. Deliver a candidate file with every source and boundary value, not just a geometry plot.

P3. Audit the kinetic equations and conservation

Formalize (3.2), its invariants, the Maxwell constraints and (5.8)–(5.9). Check velocity singularities, boundary terms and integrability. Implement reaction/radiation moments with one shared energy ledger. Reject fixtures with charge-unbalanced injection, alpha heat counted twice, or neutron energy counted as direct plasma self-heating.

P4. Establish a trusted nonlinear turbulence calculation

Reproduce a published kinetic-electron benchmark in at least one established gyrokinetic code, using a second code or published converged results for comparison. Then use the candidate's actual geometry and gradients. Keep symmetry-breaking and zonal modes. Test resolution, electromagnetic effects, domain size, kinetic species, collision operators and averaging windows. Record particle and energy fluxes with uncertainties. A linear stability scan or a fitted diffusivity is not this deliverable.

P5. Repair and calibrate geometric turbulence proxies

Resolve the physical formula and sign conventions in the existing available-energy module directly against [17]. Prove the bounce-action differentiation with its moving turning points and integrable endpoint singularities. Treat the resulting quantity as a proxy until a flux theorem or a clearly delimited empirical envelope is supplied. Compare it against P4 across both favourable and unfavourable perturbations of the same design.

P6. Search for coercivity and an observable estimate

Start with the exact selected linearization and its null/unstable modes. Try a gyrokinetic generalized free energy, relative entropy, hypocoercive cross terms and geometric commutators. Use (8.5) to identify the precise field-gradient norm required. Attempt a quantitative nonlinear estimate around P2, keeping source, collision, reference and boundary terms. If it fails, give the explicit uncontrolled term or unstable mode and test an alternative functional; do not substitute an assumed coercivity constant.

P7. Build the verified evolution bridge

For the simplest physically relevant pilot, prove the PDE-to-scalar estimate in Target 2. Use interval-enclosed continuum reconstruction, velocity-tail estimates and compatible field discretization. Improve the useful certified horizon or perturbation radius, not merely the number of formalized comparison lemmas. Repeated finite-time certificates require uniform margins or an invariant-set argument before an all-time claim.

P8. Couple transport, heating, fueling and exhaust

Use P4/P6/P7 flux information in (5.12) or in a direct moment bound. Include the actual alpha deposition fraction, radiation, impurity/ash evolution and edge matching. Establish local as well as integrated temperature margins and a kinetic fusion-rate bound. Distinguish plasma gain, wall loading and plant gain. Diagnose thermal runaway as well as cooling.

P9. Explore the complex/gauge and GR routes with a deliverable

For the complex route, deliver a reconstructed Euclidean field, a QS obstruction/compatibility theorem, or sharper coil certificate constants. For the GR route, identify the characteristic/cancellation structure and replace any unavailable dispersive decay in the torus. For Wick rotation, state and prove the real-time reconstruction theorem before using it for stability. A formal analogy without such an output is not a completed package.

P10. Independent adversarial audit

Check that all results concern the same candidate and parameter set. Change the perturbation outside the ansatz. Test a colder reference, a non-Maxwellian tail, a current error, a nonzero wall flux and an artificial-dissipation reduction. Verify that numerical convergence, a Lean proof and empirical validation are labelled separately. A theorem with its desired conclusion stored inside a structure is not a producer theorem.

12 Acceptance criteria and a minimal first milestone

The smallest useful first milestone is *not* a new all-purpose plasma solver. It is:

1. One realizable coil candidate and one explicitly delimited plasma regime.
2. A self-consistent reference with a documented continuum residual.
3. A nonlinear kinetic turbulence calculation with a credible uncertainty budget.
4. One rigorous kinetic-energy/field or flux-estimate pilot that improves the usable operating margin.

5. A coupled finite-time thermal/fuel/alpha/radiation balance on that same candidate.

At each stage state which claim has actually been achieved:

Claim	Necessary evidence
Coil realizability	Actual closed finite windings, engineering bounds and a certified field error.
Magnetic confinement	Surfaces/field topology for the actual field, plus orbit-model applicability and particle-loss estimates.
Kinetic stability	A norm/observable bound for the permitted perturbations of a specified coupled equation.
Temperature maintenance	Local or stated averaged temperature bounds with transport, deposition, radiation, fueling and exhaust included.
Fusion performance	Reaction-rate/deposition bounds, gain definition and the relevant input powers.
Physical reliability	Regime justification, benchmark/experimental evidence and an honest model-discrepancy allowance.

Three useful rejection tests are immediate. A scalar entropy can remain small while a boundary layer makes wall heat flux large. A perfectly symmetric steady ansatz can be unstable to a nonsymmetric mode. A vacuum field can have excellent guiding-centre confinement while finite-beta collective modes or electron-scale turbulence spoil thermal performance. The proposed contracts must detect all three.

The best-supported first mathematical investment is a combination of a coil/background certificate, a kinetic free-energy or relative-entropy estimate, and an a posteriori observable bound. Complex geometry can improve the background construction and the estimate. BKM-type continuation supplies the final lifespan bridge only after its hypotheses are produced. The GR analogy has a concrete MHD precedent. A direct Wick-rotation proof of a maintained kinetic fusion state remains speculative.

A Closing the reaction/data specification

Symbols for atomic and radiation physics must not conceal arbitrary favourable source terms. One exact way to specify a reaction module is as follows. Index each retained channel by r, with incoming states (a_i, z_i) and outgoing states (b_j, z'_j). States include velocity and, where needed, atomic level, photon frequency/direction or nuclear identity. A nonnegative transition measure K_r and the appropriate incoming occupation factors define the weak operator:

$$\sum_s \int \zeta_s(z)\mathcal{R}_s^{(r)}(z)\,\mathrm{d}z = \int \left[\sum_j \zeta_{b_j}(z'_j) - \sum_i \zeta_{a_i}(z_i)\right] \prod_i F_{a_i}(z_i)\,\mathsf{K}_r(\,\mathrm{d}z' \mid z)\,\mathrm{d}z. \tag{A.1}$$

Relative-speed factors, identical-particle combinatorial factors and angular differential cross-sections belong in K_r. Stimulated radiation or other occupation corrections must be specified if relevant; they are not inferred from this classical product shorthand. For photons in the intensity convention above, the number density per frequency and solid angle is $I_{\nu,p}/(ch\nu)$. Reverse reaction channels are separate channels.

Require the support and weights of each kernel to implement the relevant charge, momentum and total-energy conservation, including internal and nuclear rest energies. Taking the corresponding test functions in (A.1) proves the reaction balance identities. Consistent radiation emission/absorption

terms and matter recoil/energy terms can then be extracted from the same kernel. An energy moment alone does not determine the full operator.

The resulting PDE family becomes a particular initial-boundary-value problem only after the following are fixed:

- Coil paths/current shapes, currents or voltages, conductor properties, vessel geometry and external field/radiation boundary conditions.
- Initial distributions, compatible Maxwell fields, source spectra and controller laws.
- Species/atomic-state list, every retained reaction kernel, Coulomb-logarithm convention and a quantified omission list.
- Wall reflection/recycling/emission and radiation reflection kernels, sheath treatment and coolant conditions if wall temperature is claimed.
- If using reduced equations, their exact source/version, normalization, ordering, interfaces, and numerical/model error targets.

Data uncertainty is physical input uncertainty. It cannot be eliminated by proving the algebra of the chosen PDE. This brief supplies the equations and the required closure interfaces, not device-specific cross-section files, controls or an already validated parameter set.

B Sanity checks for the first mathematical implementation

1. **Landau masses/signs:** check $U(z)z = 0$ and $a \cdot U(z)a \geq 0$; pair the ss' and $s's$ terms to recover (3.4) and (3.5). Test unequal electron/ion masses.
2. **Fields:** with no particles and static closed currents, recover (2.1) away from conductors. With charge-conserving kinetic evolution, recover (2.8).
3. **Energy:** cancel $\mathbf{j}_{\mathrm{pl}} \cdot \mathbf{E}$ exactly. Track the work of external current supplies. Test a closed no-source system and an absorbing-wall system separately.
4. **Fusion:** one D–T event consumes two fuel nuclei, produces one helium nucleus and one neutron, preserves charge, and releases the correct nuclear energy. Test alpha escape before slowing.
5. **Geometry:** in the orientation of (7.3), verify the Hodge-star sign; verify $\mathbf{B} \times \mathbf{u} = \nabla\Psi$ from the reconstruction formula. Never replace the physical Hodge star unnoticed.
6. **Ansatz:** substitute the proposed reference into every equation. Use the nonconstant-Maxwellian calculation in Section 7 as a negative fixture.
7. **Thermal inference:** reject an inference from total heat alone to a pointwise minimum, or from temperature alone to a non-Maxwellian reaction-rate lower bound.
8. **Turbulence:** test sensitivity to nonsymmetric modes, zonal flows and electromagnetic fluctuations. A periodic box with no exhaust must not pass a wall-loss certificate.

C Primary references and source scope

The equations and proposed deductions are written in a common notation rather than copied from any single source. The literature below supports the modelling choices and identifies existing results. Research targets in Sections 8–11 are not claimed as theorems in these references. Repository links are pinned to the inspected commit.

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